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Boesen

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(54) **SELECTIVE AUDIO ISOLATION FROM BODY GENERATED SOUND SYSTEM AND METHOD**

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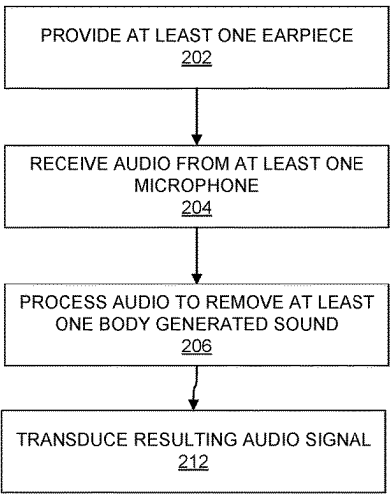
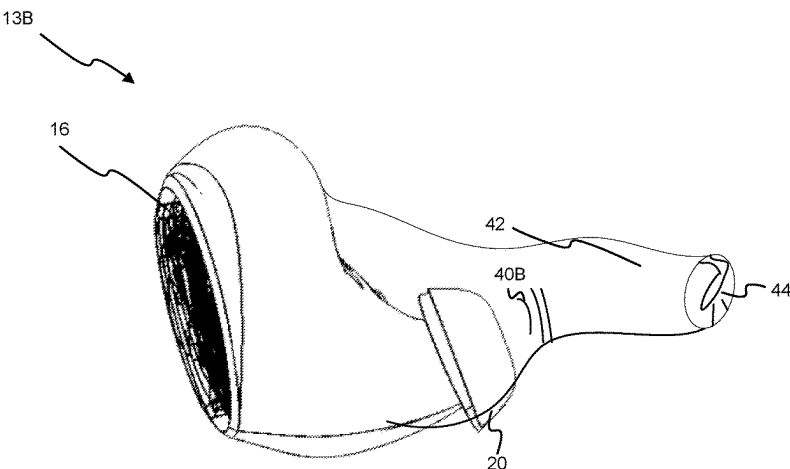
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(57) **ABSTRACT**

A wireless earpiece includes a wireless earpiece housing, a processor disposed within the wireless earpiece housing, at least one microphone operatively connected to the processor, and at least one speaker operatively connected to the processor. The processor is configured to receive audio from the at least one microphone, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker. The processing of the audio involves identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds.

16 Claims, 6 Drawing Sheets



Related U.S. Application Data

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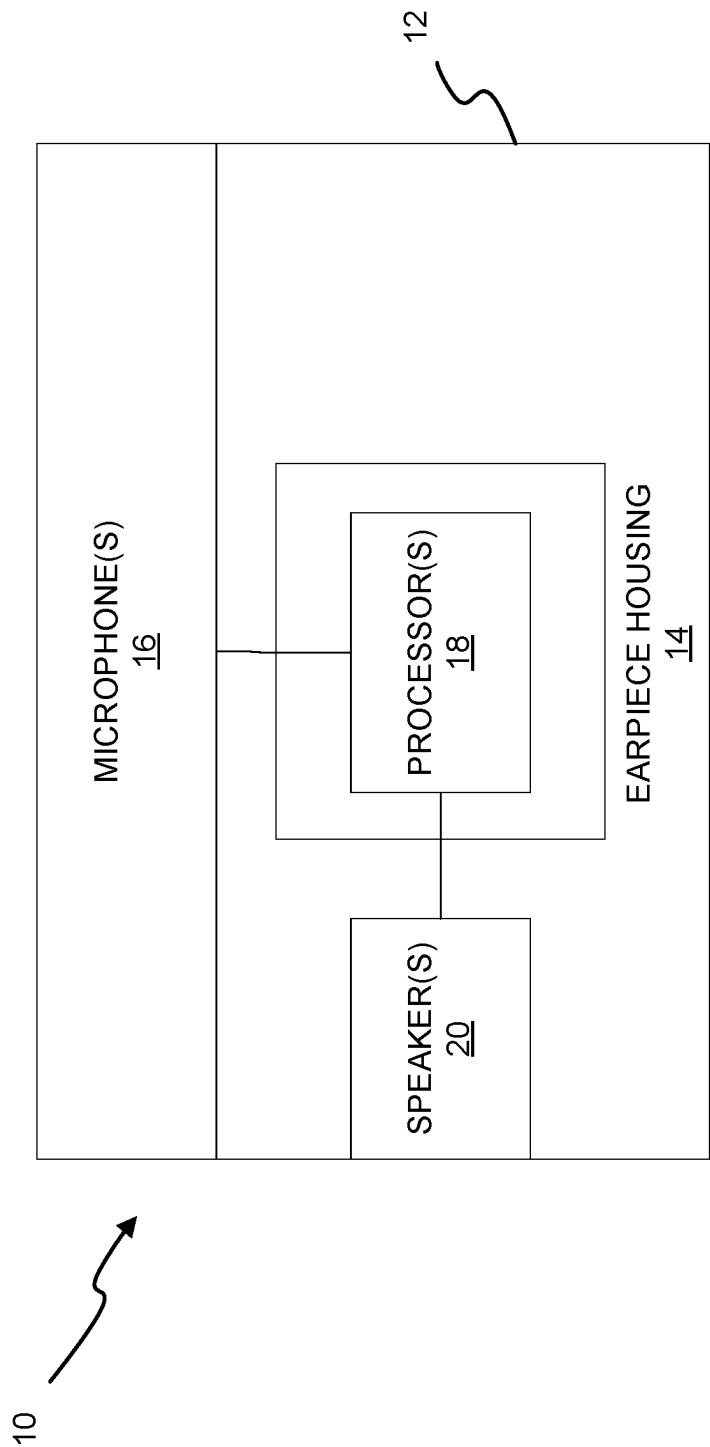


FIG. 1

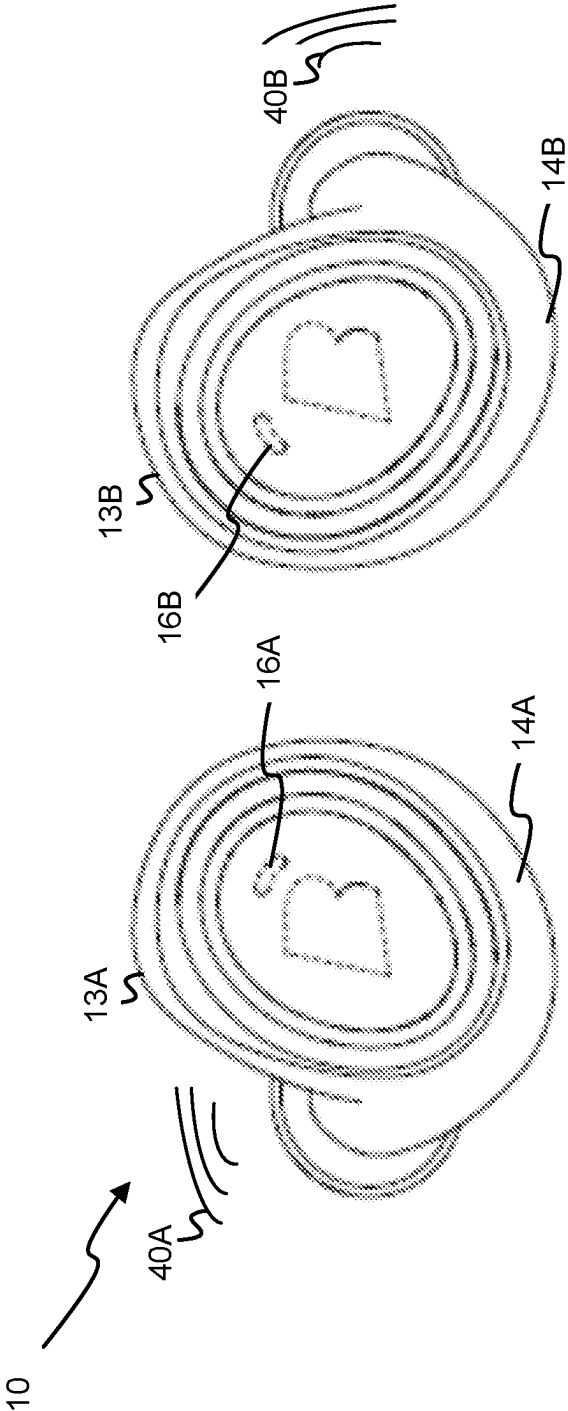


FIG. 2

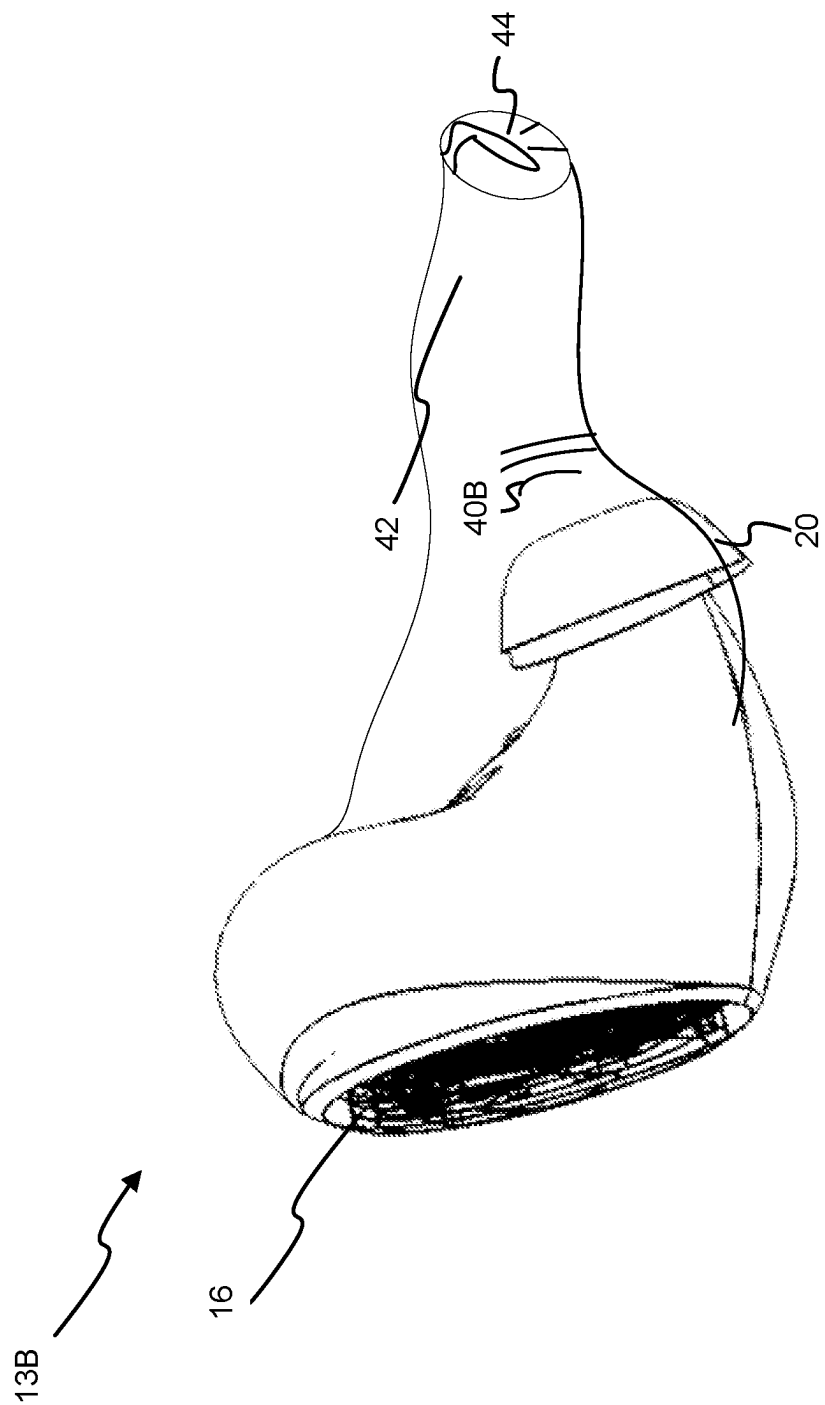


FIG. 3

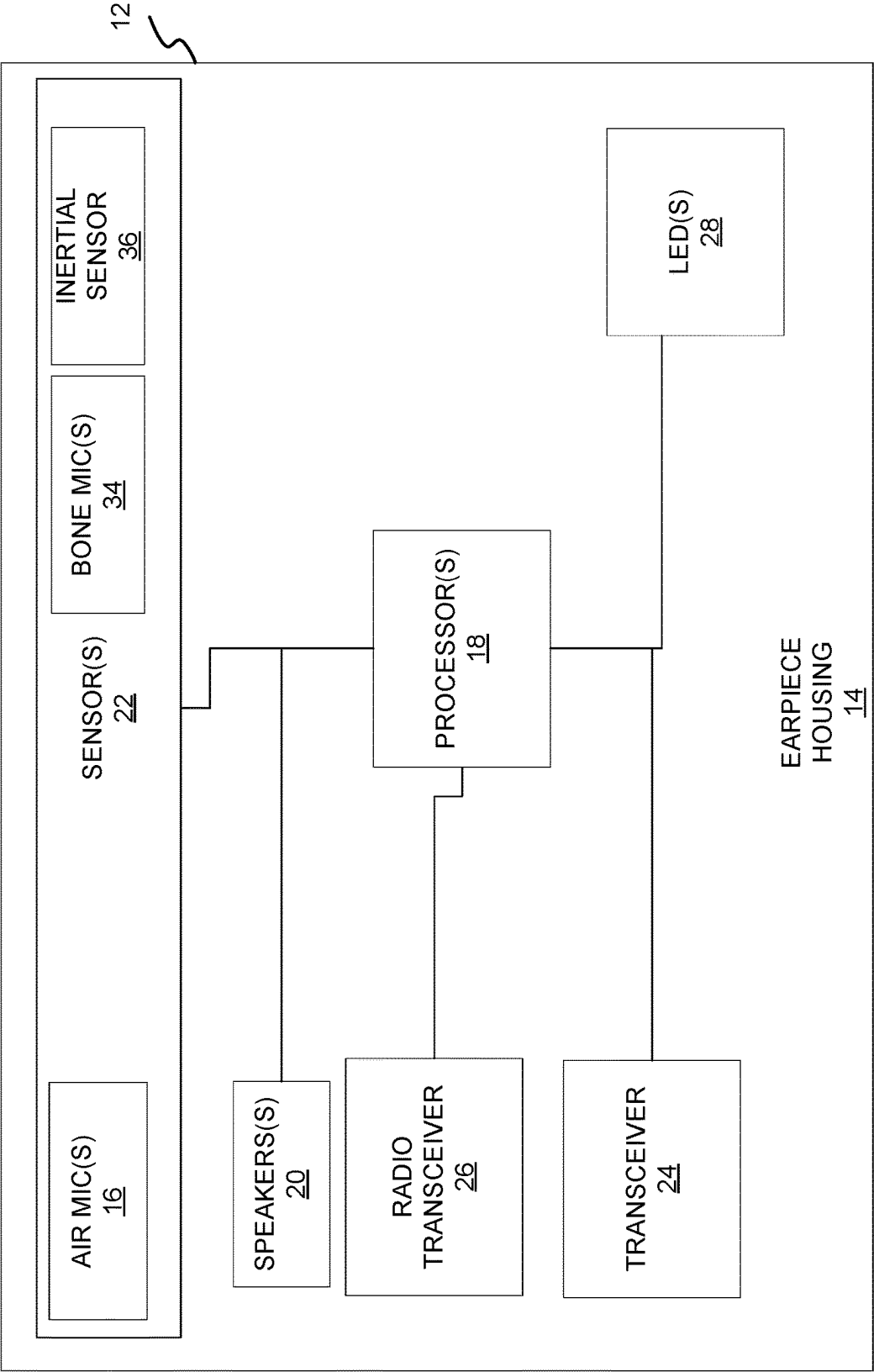


FIG. 4

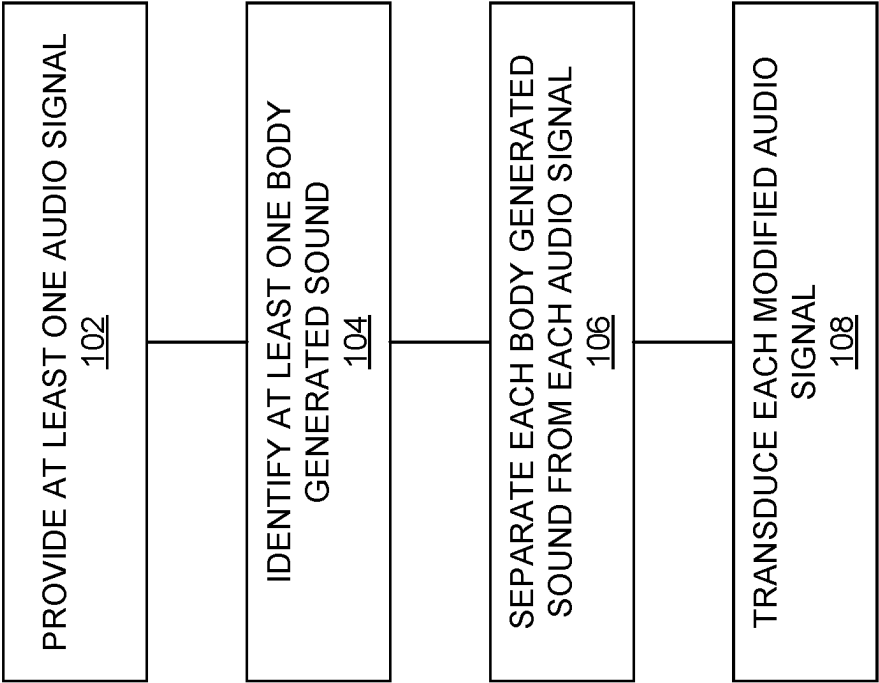


FIG. 5

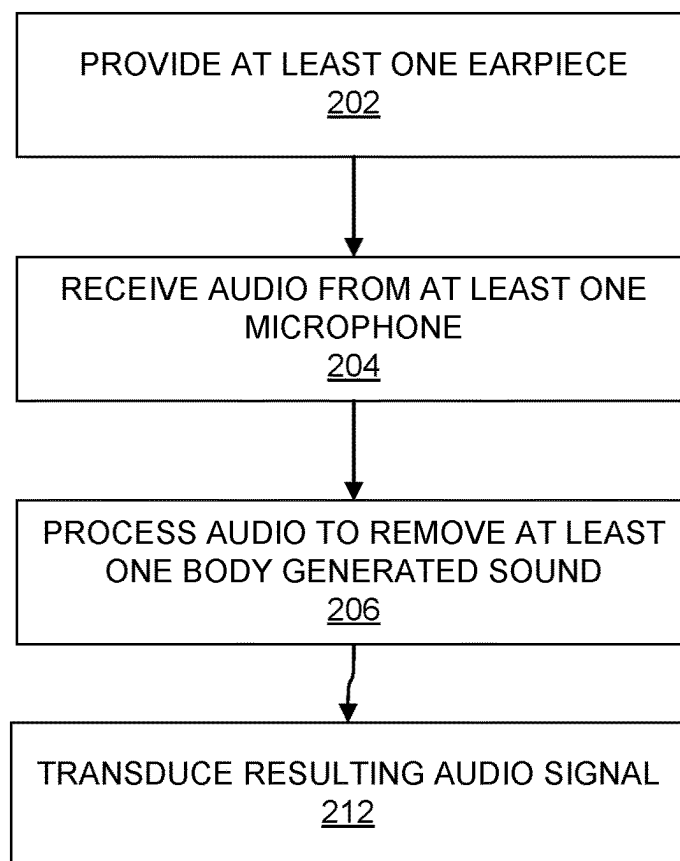


FIG. 6

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SELECTIVE AUDIO ISOLATION FROM BODY GENERATED SOUND SYSTEM AND METHOD

PRIORITY STATEMENT

This application is a continuation of U.S. patent application Ser. No. 17/879,319 filed Aug. 2, 2022, which is a continuation of U.S. patent application Ser. No. 17/147,329, filed Jan. 12, 2021 now U.S. Pat. No. 11,417,307, which is a continuation of U.S. patent application Ser. No. 16/101,894, filed Aug. 13, 2018 now U.S. Pat. No. 10,896,665, which is a continuation of U.S. patent application Ser. No. 15/801,045 filed on Nov. 1, 2017 now U.S. Pat. No. 10,062,373, which claims priority to U.S. Provisional Patent Application 62/417,195, filed on Nov. 3, 2016, all of which are titled Selective Audio Isolation from Body Generated Sound System and Method, and all of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

The present invention relates to wearable devices. More particularly, but not exclusively, the present invention relates to earpieces.

BACKGROUND

People generate a great deal of body noise through their everyday movements. While such sounds are generally not a problem, when a user wears an earpiece, the user tends to focus on such sounds to the detriment of their use and enjoyment of the earpiece. What is needed is a system and method to reduce or eliminate such body noise, so a user may enjoy the use and enjoyment of the earpiece without undue distraction.

SUMMARY

Therefore, it is a primary object, feature, or advantage of the present invention to improve over the state of the art.

It is a further object, feature, or advantage of the present invention to eliminate body noise through the use of an active noise cancellation system.

It is a still further object, feature, or advantage of the present invention to eliminate body noise through the filtering of body noise acquired using one or more microphones connected to an earpiece.

Another object, feature, or advantage is to eliminate outside noise in addition to body noise.

In one implementation, a system includes an earpiece having an earpiece housing, at least one microphone mounted onto the earpiece housing, wherein at least one microphone is configured to receive at least one body generated sound, a processor disposed within the earpiece housing and operatively connected to each microphone, wherein the processor is configured to neutralize each body generated sound, and a speaker operatively connected to the processor, wherein the speaker is configured to transduce audio signals.

One or more of the following features may be included. The earpiece housing may be composed of soundproof materials. The earpiece may comprise a set of earpieces. The set of earpieces may comprise a left earpiece and a right earpiece. Each earpiece may be configured to substantially encompass an external auditory canal of a user. Each earpiece may be further configured to substantially fit within a

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user's ear canal. One or more microphones may comprise an air microphone and a bone conduction microphone. One or more microphones may be further configured to filter one or more body generated sounds. The body generated sound filtering may be performed within a specific decibel range. The neutralization of each body generated sound may be performed by superimposing a cancellation signal to an audio signal communicated by the speaker, wherein the cancellation signal is configured so the superimposition of each cancellation signal with each body generated noise substantially nets to zero decibels. The neutralization of each body generated sound may be performed by filtering each body generated sound from audio signals using one or more microphones. Audio signals to the speaker may be suppressed or attenuated if the audio signal exceeds a specific decibel range.

In another implementation, a method of filtering body generated sounds from an audio signal using an earpiece includes providing at least one audio signal, receiving at least one body generated sound through a microphone, separating each body generated sound from each audio signal using a processor, and communicating each audio signal using a speaker.

One or more of the following features may be included.

One or more earpieces may be configured to substantially encompass an external auditory canal of a user. One or more microphones may filter one or more audio signals. The separation of each body generated sound from each audio signal may be performed through the generation of a cancellation signal by the processor, wherein the cancellation signal is configured as to substantially net to zero decibels when superimposed onto a body generated sound. The separation of each body generated sound from each audio signal may be performed through the filtering of each body generation sound.

According to another aspect, a wireless earpiece includes a wireless earpiece housing, a processor disposed within the wireless earpiece housing, at least one microphone operatively connected to the processor, and at least one speaker operatively connected to the processor. The processor is configured to receive audio from the at least one microphone, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker. The processing of the audio may include identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds. The at least one microphone may include both an air microphone and a bone microphone. The identifying the body generated sounds may be performed by comparing a first audio signal from the air microphone with a second audio signal from the bone microphone. The removing the body generated sounds may be performed by superimposing a cancellation signal to the audio. The removing the body generated sounds may be performed by filtering the body generated sounds from the audio. The body generated sounds may include at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

According to another aspect, a method of removing body generated sounds by an earpiece may include providing the earpiece. The earpiece may include a wireless earpiece housing, a processor disposed within the wireless earpiece housing, at least one microphone operatively connected to the processor and at least one speaker operatively connected to the processor, wherein the processor is configured to receive audio from the at least one microphone, perform processing on the audio to provide processed audio, and output the processed audio to the at least one speaker. The

method may include receiving audio from the at least one microphone at the processor, performing processing on the audio to provide processed audio, wherein the processing of the audio comprises identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds, and outputting the processed audio to the at least one speaker of the wireless earpiece. The at least one microphone may include an air microphone and a bone microphone. The step of identifying the body generated sounds may be performed by comparing a first audio signal from the air microphone with a second audio signal from the bone microphone. The step of removing the body generated sounds may be performed using active noise cancellation. The step of removing the body generated sounds may be performed by filtering.

According to another aspect, a set of wireless earpieces includes a first wireless earpiece having a first wireless earpiece housing, a processor disposed within the first wireless earpiece housing, at least one first wireless earpiece microphone operatively connected to the first wireless earpiece processor, a first wireless earpiece transceiver operatively connected to the first wireless earpiece processor, and at least one first wireless earpiece speaker operatively connected to the first wireless earpiece processor. The set of wireless earpieces may further include a second wireless earpiece comprising a second wireless earpiece housing different from the first wireless earpiece housing, at least one second wireless earpiece microphone, at least one second wireless earpiece speaker, and at least one second wireless earpiece transceiver. The transceiver of the second wireless earpiece may communicate with the transceiver of the first wireless earpiece. The first wireless earpiece processor may be configured to receive audio from the at least one first wireless earpiece microphone and from the at least one second wireless earpiece microphone, perform processing of the audio to provide processed audio, and output the processed audio to the at least one first wireless earpiece speaker. The processing of the audio may involve identifying body generated sounds generated by a body of a user of the first wireless earpiece and the second wireless earpiece and removing the body generated sounds. The at least one first wireless earpiece microphone may include a first wireless earpiece air microphone and a first wireless earpiece bone microphone. The at least one second wireless earpiece microphone may include a second wireless earpiece air microphone and a second wireless earpiece bone microphone. The step of identifying the body generated sounds may be performed by comparing audio signals from the first wireless earpiece air microphone with the first wireless earpiece bone microphone and from the second wireless earpiece air microphone with the second wireless earpiece bone microphone. The step of removing the body generated sounds may be performed by superimposing a cancellation signal to the audio. The step of removing the body generated sounds may be performed by filtering the body generated sounds from the audio. The body generated sounds may include at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

One or more of these and/or other objects, features, or advantages of the present invention will become apparent from the specification and following claims. No single embodiment need provide each and every object, feature, or advantage. Different embodiments may have different objects, features, or advantages. Therefore, the present invention is not to be limited to or by an object, feature, or advantage stated herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram regarding a first embodiment of the system.

FIG. 2 illustrates a left earpiece and a right earpiece.

FIG. 3 illustrates a side view of the right earpiece and its relationship with a user's ear.

FIG. 4 is another block diagram regarding a second embodiment of the system.

FIG. 5 includes a flowchart of one implementation of the method of filtering body generated sounds from an audio signal using an earpiece.

FIG. 6 includes a flowchart of a second implementation of the method of filtering body generated sounds from an audio signal using an earpiece.

DETAILED DESCRIPTION

FIG. 1 shows a block diagram of one embodiment of the system 10 having an earpiece 12 with an earpiece housing 14, one or more microphones 16 associated with the earpiece housing 14, a processor 18 disposed within the earpiece housing 14 and operatively connected to each microphone 16, and a speaker 20 disposed within the earpiece housing 14 and operatively connected to the processor 18. The earpiece housing 14 may be composed of any material resistant to shear and strain and unlikely to cause skin allergies or rashes. In addition, the earpiece housing 14 may be composed of soundproof materials and may also be configured as to be waterproof. The earpiece housing 14 may also substantially encompass the external auditory canal of a user in order to substantially reduce or eliminate external sounds. One or more microphones 16 associated with the external housing 14 may be configured to receive external sounds in addition to body generated sounds.

One or more microphones 16 may be located on or within any part of the earpiece housing 14 conducive to receiving a body generated sound and may also include air and bone conduction microphones configured to receive sounds. In one mode of operation, the wireless earpiece may prevent environmental sounds from being heard by a user. In another mode of operation, the wireless earpiece 12 may provide an audio transparency function where sounds detected from the environment with one or more microphones 16 may be reproduced at one or more speakers 20 of the wireless earpiece 12.

Various methods of identifying body generated sounds are contemplated. In one embodiment audio signals from one or more bone conduction microphones are compared with audio signals from one or more air conduction microphones to identify body generated sounds. In another embodiment, processing of audio signals may be used to determine a source of the sound, for example when two microphones of a wireless earpiece detect the same sound at different instances in time, a determination may be made if the sound propagated from the environment to the microphones or from the body to the microphones. In addition, body generated sounds may be determined by comparison of audio to known body generated sounds or comparison of properties of audio signals to properties of known body generated sounds. Such comparisons may be facilitated through the use of training sets of body generated sounds or asking a user to generate different body generated sounds. Any number of other methods may be used to identify as sound as a body generated sound. The processor 18 is disposed within the earpiece housing 14 and operatively connected to each microphone 16 and is configured to neutralize or otherwise

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attenuate or cancel any body generated sound signal it receives. The processor 18 may perform this function through active noise cancellation, where a cancellation signal is superimposed onto each body generated sound through destructive interference so as to substantially reduce the effect of the body generated sound, or may simply filter out a body generated sound using an algorithm stored within the processor 18 configured to identify which sounds may be body generated sounds identified by the algorithm from reaching the processor 18. These processes may be performed continuously or discretely, and the processor 18 may determine more than one cancellation signal if necessary depending on the body generated sounds received by the microphones 16. The processor 18 may also be configured to produce the audio signals communicated to a user's tympanic membrane, filter out external sounds which are not body generated sounds, and/or regulate other functions of the earpiece 12. The speaker 20 configured to transduce audio signals substantially free of body generated sounds to a user's tympanic membrane.

FIG. 2 illustrates a system 10 which includes a left earpiece 13A and a right earpiece 13B. The left earpiece 13A has a left earpiece housing 14A. The right earpiece 13B has a right earpiece housing 14B, the left earpiece housing 14A different from the right earpiece housing 14. The left earpiece 13A and the right earpiece 13B may be configured to fit around a user's ear canal so as to minimize the amount of external sound capable of reaching the ear canal as well as configured to fit within the ear canal so as to minimize the distance between the speakers and a user's tympanic membranes and maintain good fit. The earpiece housings 14A and 14B may be composed of soundproof or shear resistant materials and may also be configured to be waterproof. A microphone 16A is shown on the left earpiece 13A and a microphone 16B is shown on the right earpiece 13B. The microphones 16A and 16B may be located anywhere on the earpieces 13A and 13B respectively and may be configured to receive ambient environmental sounds or a user's instructions to control one or more earpiece functions in addition to non-vocal body generated sounds. The earpieces 13A and 13B may be configured to communicate audio signals 40A and 40B respectively to a user's tympanic membranes.

FIG. 3 illustrates a side view of a right earpiece 13B and its relationship to a user's ear. The right earpiece 13B may be configured to both minimize the amount of external sound reaching the user's ear canal 42 and to facilitate the transmission of the audio signal 40B from the speaker 20 to a user's tympanic membrane 44. The right earpiece 13B may be configured to be of any size necessary to fit within the user's ear canal 42 and the distance between the speaker 20 and the user's tympanic membrane 44 may be any distance sufficient to facilitate transmission of the audio signal 40B to the user's tympanic membrane 44. A microphone 16 is shown on the outside of the right earpiece 13B. The microphone 16 may be configured to, in addition to receiving body generated noise, receive sounds from a user or the outside environment which may be used to reconfigure, change, or otherwise modify one or more functions of the earpiece 13B.

FIG. 4 is a block diagram of another embodiment of the system which includes an earpiece 12 with an earpiece housing 14, a plurality of sensors which may include an air microphone 16 and a bone microphone 34. Additional microphones may be present. A speaker 20 is operatively connected to the processor 18. Other examples of sensors 22 which may be present may include one or more inertial sensors 36. A radio transceiver 26 is operatively connected to the one or more processors 18. Another transceiver 24

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may be present which may provide for communicating between left and right earpieces. The transceiver 24 may be a near field magnetic induction (NFI) transceiver. One or more LEDs may also be operatively connected to the one or more processors 18 to provide for visual feedback of operations of the device.

FIG. 5 illustrates one implementation of the method of removing body generated sounds. In step 102, an audio signal is provided such as may be detected using one or more microphones of an earpiece. In step 104, at least one body generated sound is identifying within the one or more audio signals. The body generated sounds may originate from anywhere on a user's body and need not be physiologic in nature. For example, a body generated sound may be due to a necklace moving around the user's neck, a sole installed within a shoe the user is wearing, baggy pants the user may be wearing, or otherwise be body generated. In step 106, a processor separates any audio signals from body generated sounds. The process by which a processor may perform this step include through a noise cancellation signal or through filtering or active filtering using one or more algorithms or otherwise. In step 108, the modified audio signal is transduced.

FIG. 6 shows another example of a method of filtering body generated sounds from an audio signal using an earpiece. In step 202, an earpiece is provided such as one of the examples previously described. In step 204, audio is received from one or more of the microphones. In step 206, the audio is processed to remove at least one body generated sound. In step 212, the resulting audio signal(s) are transduced at one or more speakers of the wireless earpiece.

It should also be appreciated where there is a set of wireless earpieces such as shown in FIG. 2, audio from one of the wireless earpieces may communicated to the other earpiece and be processed by the other earpiece to remove body generated sounds. Having audio from both locations may provide additional information about the sound to assist in processing, for example, it may provide additional information regarding locating the sound.

Thus, the system and method described herein allow for selectively monitoring for body generated sounds, e.g. the click or pop generated by movement of the user's temporomandibular joint. Such sounds may be detected by microphones incorporated into the earpiece device and then extracted from the audio output signal delivered through the speaker system of the device to the middle ear. Such a device allows for mitigating the effects of body induced sound in order to allow the user to concentrate more fully on the selected device output. Moreover, an earpiece as described herein allows for monitoring body created sound inputs delivered to the middle ear through the air or bone conduction systems present in the body, allows for creating effective audio isolation through the use of isolation measures which would include masking noise or signal extraction algorithms, and allows a user to more fully concentrate on the audio input delivered to the middle ear by reducing or subtracting the sound inputs from body generated noise inputs.

Therefore, various methods, system, and apparatus have been shown and described. Although specific examples or embodiments are shown herein it is to be understood various elements or steps from different embodiments may be combined. It is to be further understood numerous options, variations, and alternatives are contemplated.

What is claimed is:

1. A wireless earpiece comprising: a wireless earpiece housing; a processor disposed within the wireless earpiece

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housing; a plurality of microphones operatively connected to the processor; at least one speaker operatively connected to the processor; wherein the processor is configured to receive audio from the plurality of microphones, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker to provide an audio transparency function; wherein the processing of the audio comprises identifying sounds to remove based on training data associated with sounds to remove and then remove the sounds to remove from the processed audio, wherein the sounds to remove are body generated sounds generated by a body of a user of the wireless earpiece.

2. The wireless earpiece of claim 1, wherein the removing the body generated sounds is performed by superimposing a cancellation signal to the audio.

3. The wireless earpiece of claim 1, wherein the removing the body generated sounds is performed by filtering the body generated sounds from the audio.

4. The wireless earpiece of claim 1, wherein the body generated sounds comprise at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

5. The wireless earpiece of claim 1, wherein the body generated sounds comprise sounds generated by movement of the user.

6. The wireless earpiece of claim 1, wherein the body generated sound comprises sounds generated by items worn by the user as the user moves.

7. The wireless earpiece of claim 1 wherein the processor is further configured to receive the audio from one or more microphones of a second wireless earpieces in operative communication with the wireless earpiece.

8. A method of removing body generated sounds by an earpiece comprising:

providing the earpiece, the earpiece comprising a wireless earpiece housing, a processor disposed within the wireless earpiece housing, a plurality of microphones operatively connected to the processor and at least one speaker operatively connected to the processor, wherein the processor is configured to receive audio from the plurality of microphones, perform processing on the audio to provide processed audio, and output the processed audio to the at least one speaker;

receiving audio from the plurality of microphones at the processor;

performing processing on the audio to provide processed audio, wherein the processing of the audio comprises

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identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds from environmental audio detected by the plurality of microphones based on training data associated with body generated sounds; and

outputting the processed audio to the at least one speaker of the wireless earpiece.

9. The method of claim 8 wherein the plurality of microphones include an air microphone and a bone microphone.

10. The method of claim 8 wherein the removing the body generated sounds is performed using active noise cancellation.

11. The method of claim 8 wherein the removing the body generated sounds is performed by filtering.

12. The method of claim 8 wherein the body generated sounds comprise at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

13. The method of claim 8 wherein the body generated sounds comprise sounds generated by movement of the user.

14. The method of claim 8 wherein the body generated sound comprises sounds generated by items worn by the user as the user moves.

15. The wireless earpiece of claim 8 wherein the processor is further configured to receive the audio from one or more microphones of a second wireless earpieces in operative communication with the wireless earpiece.

16. A wireless earpiece comprising:

a wireless earpiece housing;

a processor disposed within the wireless earpiece housing;

a plurality of microphones operatively connected to the processor;

at least one speaker operatively connected to the processor;

wherein the processor is configured to receive audio from the plurality of microphones, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker to provide an audio transparency function;

wherein the processing of the audio comprises identifying body generated sounds generated by a body of a user of the wireless earpiece to remove based on training data associated with sounds to remove and then remove the sounds to remove from the processed audio.

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